

BST Physics Curriculum Vol 11 Chapter 1 — Bounded Hermitian Symmetric Domains v0.4 SIGNATURE (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

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Vol 11 Chapter 1 — Bounded Hermitian Symmetric Domains

Chapter motivation

Cartan classified bounded Hermitian symmetric domains (HSDs) in 1894-1935; Helgason 1978 Theorem X.6.1 provides modern presentation. There are exactly 6 irreducible types: 4 infinite families (Types I-IV) + 2 exceptional (Types E_6 + E_7). BST's substrate is $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ (Type IV, complex dimension $n_C = 5$). The Strong-Uniqueness Theorem (Vol 0 Ch 9) selects D_{IV}^5 uniquely among all HSDs via 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 authoritative enumeration.

Section 1.0b — Reader-grade 3-level pedagogy

Level 1: Cartan 1894 + Helgason 1978 X.6.1: 6 irreducible bounded HSD types (4 infinite families I-IV + 2 exceptional E_6 , E_7); BST substrate $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ uniquely selected by Strong-Uniqueness Theorem 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99.

Level 2 (graduate): Cartan 1894-1935 classified Riemannian symmetric spaces; Helgason 1978 *Differential Geometry, Lie Groups, and Symmetric Spaces* (Academic Press) Theorem X.6.1 gives the 6 irreducible bounded HSD types: - Type I_{p,q}: $SU(p,q)/[SU(p) \times SU(q) \times U(1)]$ (Grassmannians of complex p-planes in $\mathbb{C}^{p+q}$); complex dim = p·q; rank = min(p, q) - Type II_n: $SO^*(2n)/U(n)$; complex dim = $n(n-1)/2$; rank = $\lfloor n/2 \rfloor$ - Type III_n: $Sp(n, \mathbb{R})/U(n)$; complex dim = $n(n+1)/2$; rank = n - Type IV_n: $SO_0(2, n)/[SO(2) \times SO(n)]$ (Lie ball; quadric in projective space); complex dim = n; rank = 2 for all $n \geq 2$ - Type E_6 : $E_{6(-14)}/[SO(10) \times T]$ complex

dim 16 rank 2 - Type E_7: $E_{\{7(-25)\}}/[E_6 \times T]$ complex dim 27 rank 3

Type IV unique infinite family with rank = 2 at all $n \geq 2$: T1925 Argument B + Strong-Uniqueness C1 (T2443 RIGOROUSLY CLOSED) anchors this property.

BST substrate selection: D_{IV}^5 at $n_C = 5$ satisfies 11 RIGOROUSLY CLOSED Strong-Uniqueness criteria (Vol 0 Ch 9): C1 rank=2 + C2 $N_c=3$ + C3 $n_C=5$ + C4 $C_2=6$ + C5 $g=7$ + C6 $N_{\max}=137$ + C7-C13 substrate-Hilbert space + Bridge Object + operator zoo + Q-cluster + 4-Zone + Casimir. Null-model $\leq (1/3)^{19} \approx 8.6 \times 10^{-10}$. Per Cal #99 authoritative enumeration: 7 additional candidate criteria C7+C9+C15+C16+C17a+C17b+C18; if all ratify, null-model $\leq (1/3)^{26} \approx 3.9 \times 10^{-13}$.

Level 3 (5th-grader): A “Hermitian symmetric domain” is a special bounded curved complex space with a lot of symmetry. Cartan classified all of them in 1894 — there are exactly 6 types (4 infinite families + 2 special). BST’s substrate is the Type IV one with complex dimension 5: D_{IV}^5 . Why this one specifically? The Strong-Uniqueness Theorem (Vol 0 Ch 9) proves D_{IV}^5 is uniquely picked out by 11 independent criteria. The probability that all 11 criteria would align on the wrong space by accident is less than 1-in-a-billion (probability $\leq (1/3)^{19} \approx 10^{-9}$).

Section 1.1 — Cartan 1894 Classification

Six irreducible types I-IV + E_6 + E_7 with explicit Lie-group quotient realizations.

Section 1.2 — Helgason 1978 X.6.1

Modern presentation; coset realization G/K with G semisimple Lie group, K maximal compact subgroup, K -types via highest weights.

Section 1.3 — Type IV Unique Infinite Family with rank = 2

Per T1925 Argument B + T2443 Strong-Uniqueness C1: Type IV is unique infinite family with rank = 2 at all complex dimensions $n \geq 2$.

Section 1.4 — BST Substrate Selection

D_{IV}^5 at $n_C = 5$ + rank = 2 satisfies Strong-Uniqueness 11 RIGOROUSLY CLOSED + 7 candidates per Cal #99 (Vol 0 Ch 9 v0.8 + Saturday Wave 1 cross-references).

Section 1.5 — Hua 1958 Coordinate Realization

Type IV bounded domain realization: $D_{IV}^5 \subset \mathbb{C}^5$ as set of $z = (z_1, \dots, z_5) \in \mathbb{C}^5$ satisfying $1 - 2(\bar{z} \cdot z) + |z \cdot z|^2 > 0$ + $|z \cdot z| < 1$. Canonical realization per Cartan 1894 + Hua 1958.

Section 1.6 — Honest scope

- Cartan classification standard [\[?\]](#)
- D_{IV}^5 selection via Strong-Uniqueness (Vol 0 Ch 9)
- Multi-criterion convergence with null-model bounds
- Alternative HSDs structurally ruled out at if-and-only-if level on 11 criteria

Section 1.7 — Connection

- Vol 0 Ch 1 D_{IV}^5 APG
- Vol 0 Ch 9 Strong-Uniqueness Theorem
- Vol 1 Ch 2 Substrate Hilbert Space
- Vol 11 Ch 5 D_{IV}^5 Geometry: Coset + Boundary + Coordinates (deeper coordinate treatment)

— Lyra, Vol 11 Ch 1 v0.3 SIGNATURE chapter, Saturday 2026-05-23 EDT